AC DRIVING, NEAR MISSES & OBSTRUCTIONS

Revision Guide — Driver Rules & Procedures

GERT8000-AC Iss 10 | GERT8000-TW1 Iss 22 | EMR PPTs 100-02, 33-01, 27-01

PART 1 — DRIVING UNDER AC ELECTRIFIED LINES

Source: GERT8000-AC Iss 10 | EMR PPT 100-02

1. OLE AND SAFETY

The Overhead Line Equipment (OLE) carries 25,000 volts AC. You must treat it as live and dangerous at all times, even after a switch-off — residual current remains.

CRITICAL: Minimum Safe Distance

You and anything you are carrying must be **no nearer than 2.75 metres (9 feet)** from live OLE, pantographs and associated roof-mounted electrical equipment.

If you need to enter this distance for any reason other than an emergency rescue, you must request the OLE is **isolated and earthed**.

Exception: if you are rescuing a person who is NOT in contact with the OLE, you may approach within 2.75m — but no nearer than 2.75m.

Source: GERT8000-AC §3.1, §7

OLE Equipment Components

Component	What It Is
Contact Wire	The wire the pantograph slides along to draw current
Catenary Wire	Upper supporting wire above the contact wire
Dropper	Hangs the contact wire from the catenary wire
OLE Stanchion	Lineside mast or structure that supports the whole OLE
Registration Arm	Positions the contact wire laterally over the track
Red Bond	Electrical connection between structure and running rail
Insulator	Prevents electrical current passing to earth through the structure

Source: GERT8000-AC Iss 10 | EMR PPT 100-02 slide 4

2. PANTOGRAPHS AND THE LINE LIGHT

Pantographs are devices on the train roof that make contact with the contact wire to draw electrical power. When a pantograph makes contact with live OLE, a **Line Light** illuminates on the driver's desk to confirm voltage is being drawn.

Line Light

The line light is an **indicator on your desk** showing that current is being drawn from the OLE.
It illuminates when the pantograph is in contact with live OLE.
If it goes out — stop and investigate using the decision procedure in Section 4.

VCB — Vacuum Circuit Breaker

Located on the train roof. When it operates it **cuts the power** from the OLE to the train.
Operated automatically by APC (Automatic Power Control) when crossing a neutral section, or when a fault occurs.
Will auto-reset when passing over the APC magnet, or when the driver presses the reset/pan-up button.

Source: GERT8000-AC Iss 10 | EMR PPT 100-02 slides 5, 8

3. NEUTRAL SECTIONS (OHNS)

A neutral section (also called an Overhead Neutral Section or OHNS) is a short dead section of OLE between two live electrical supply sections. It prevents a short circuit between the two supplies.

The train must **not draw power through the neutral section**. Power is automatically disconnected by an APC magnet which signals the on-train APC equipment to open the VCB. A second APC magnet closes it when the train has cleared the section.

Section	Length
Run-in (insulated)	100 feet / 30 m
Dead section	15 feet / 4.5 m
Run-off (insulated)	40 feet / 12 m
Boards spaced apart	Approximately 1 mile between boards

Stopped in a Neutral Section — Five Possible Actions (AC Module §)

1. Reset/close the VCB — your pantograph may not be in the dead section
2. Arrange for assistance — roll the train out of the OHNS
3. Raise another pantograph on the train
4. Change to Diesel mode on a Bi-Mode/Multi-Mode train
5. Lower all pantographs and arrange assistance (as for line blocked to electric trains)

Defective APC Inductor — Driver's Action

If told the APC track inductor is defective, you must **shut off power immediately before entering the neutral section** manually. (GERT8000-AC §18.2)

Source: GERT8000-AC §18.1–18.2 | EMR PPT 100-02 slide 9

4. ADD — AUTOMATIC DROPPING DEVICE

The ADD is a safety device that automatically lowers the pantograph if it detects an air leak in the pantograph head or excessive wear on the carbon strips.

When ADD Operates

1. The **pantograph automatically lowers**.
2. The **VCB opens** (cuts power).
3. The **line light extinguishes**.

ADD Operation — Driver's Action: DROP STOP BOX

DROP — the pantograph has already lowered automatically
STOP — stop the train (unless you can continue to a suitable location — see Section 5)
BOX — report to the signaller by REC

Isolating the ADD (GERT8000-AC §12.8)

If it becomes necessary to isolate the ADD, you must:

- Isolate the ADD as shown in the instructions for the type of traction concerned and your company instructions
- Tell the signaller
- Carry out the instructions you are given

Speed Limit After ADD Isolated

When proceeding with the affected pantograph raised after isolating the ADD, you must **not exceed 100 mph (160 km/h)** until the pantograph has been examined and the ADD reset.
You may be instructed to keep to a speed **lower than 100 mph** for some or all of the journey.

Source: GERT8000-AC §12.8 | EMR PPT 100-02 slides 14–16

5. LINE LIGHT GOES OUT — WHAT TO DO

Source: GERT8000-AC §12.1–12.3 | EMR PPT 100-02 slides 17–19

STOP IMMEDIATELY (DROP STOP BOX) if ANY of these apply

Damage to the OLE — something on the OLE that could contact a pantograph or the train; anything unusual; unusual noises or movement (§12.1a)

ADD has operated AND line light goes out — UNLESS more than one pantograph is raised AND the line light remains on (§12.1b)

Line light goes out, one reset attempt failed, AND EITHER: the only pantograph in use is NOT on one of the first three vehicles, OR there is more than one pantograph in use (§12.1c)

CAN CONTINUE TO A SUITABLE LOCATION if ALL of these apply

Line light goes out

Only ONE pantograph in use AND it is on one of the first three vehicles

ADD is not isolated and has not operated

No unusual movement or noises from the OLE

You have made one attempt to reset — not successful

Source: GERT8000-AC §12.2

CAN CONTINUE NORMALLY if ALL of these apply

ADD available but has NOT operated

No unusual movement or noises from the OLE

You can reset at first attempt, OR line light is restored

You can regain power

Source: GERT8000-AC §12.3

Also: ADD operates but line light does NOT go out

If there is more than one pantograph raised on the train and the ADD operates but the line light does not go out, you can continue to a suitable location before reporting to the signaller.

6. SEQUENTIAL TRIPPING / TRIPPING

Sequential tripping occurs when the train causes a series of electrical circuit breaks in the OLE. The signaller will tell you if you need to visually examine the OLE and all pantographs for damage.

Checking Without Leaving the Train

If you can check using a pantograph camera or CCTV and see no damage, tell the signaller.
The signaller will tell you whether you need to leave the train to check.

Before Leaving the Train

Tell the signaller whether conditions at your location are darkness or whether visibility is affected by fog, bright sunlight etc.
Check there is no damaged OLE near the door you would use to exit. If there is, **stay in the train** and tell the signaller.
The signaller will tell you when you can leave the train.

If Visibility Deteriorates During the Examination

STOP the examination.
Tell the signaller.
Wait until the signaller tells you the OLE has been switched off before continuing.
Continue to treat the OLE as live and dangerous.

If there is evidence that something other than a pantograph has been in contact with the OLE, or a pantograph is damaged, tell the signaller immediately.

Source: GERT8000-AC §12.4 | EMR PPT 100-02 slides 20–21

7. WHAT TO TELL THE SIGNALLER AFTER STOPPING

Source: GERT8000-AC §12.5

In ALL cases when you have stopped the train, you must tell the signaller:

- What has happened
- The location of the incident
- The location where the train has stopped
- The **nearest OLE structure number**, if you can see it from the train

If you can already do so, also tell the signaller:

- The extent of any damage to the OLE
- If there is any damage to a pantograph

Reporting to the ECO

You must make sure the following are reported to the **Electrical Control Officer (ECO)** immediately (via signaller if you do not normally speak directly to the ECO):

- Any broken or parted rails — do NOT touch them
- Any broken or defective bond — do NOT touch it
- Any equipment connected to the bond
- If the damage or defect will affect the safe operation of trains — report to signaller FIRST

8. EXAMINING THE OLE

Source: GERT8000-AC §14.2 | EMR PPT 100-02 slide 25

If asked to examine the OLE, you must:

- Proceed at caution and **not exceed 20 mph (30 km/h)**
- Look out for any damage or other problem with the OLE
- Be accompanied by a **competent person** during darkness, poor visibility or where there is a tunnel in the affected section

Checking If It Is Safe to Coast or Use Own Traction

The signaller may ask you to check whether it is safe for trains to coast with pantographs lowered or proceed using own traction power. Check ALL of these:

1. Any obstruction is **not more than 150 mm (6 inches)** below the contact wire
2. **Not more than two consecutive droppers** have come off
3. The object or defect is **more than FIVE OLE structures** away from a tunnel or overbridge
4. No other defect is obvious

Detained with a Damaged Pantograph (§12.7)

If the train cannot proceed due to pantograph damage but damage is not severe, the pantograph may be raised to supply electrical power (heating, lighting) while waiting for assistance.

Critical Rules When Raising Damaged Pantograph for Power

Immediately after raising — check it is correctly in contact with OLE and there is no arcing.
No movement of the train is allowed with pantograph raised.
Pantograph **must be lowered** before the assisting train is attached.

9. COASTING WITH PANTOGRAPHS LOWERED

Source: GERT8000-AC §15.4–15.5 | EMR PPT 100-02 slides 29–33

20 mph Coasting

If the signaller authorises trains to coast with pantographs lowered through an affected area at 20 mph:

- You must **not exceed 20 mph** through the affected area
- Lineside signs: Lower Pantograph sign, then Raise Pantograph sign at exit

High-Speed Coasting Signs

High-speed coasting is authorised by specific lineside boards. These signs must all be in place before you may proceed at high speed:

Sign	Purpose
Advance Lower Pantograph (with flashing white lights)	Warning 400 metres ahead — start reducing speed
Lower Pantograph	Lower pantographs now
Raise Pantograph	Safe to raise pantographs
Do Not Raise Pantograph (no AWS)	Do not raise pantographs in this area

Missing or Defective High-Speed Signs

If any board is missing or the white flashing lights on the advance warning board are defective, you must **report this to the signaller after passing through the affected area** — stopping specially to do so.

10. TRACTION CHANGEOVER

Source: GERT8000-AC §17–18 | EMR PPT 100-02 slides 34–37

A Power Changeover (PCO) is where a multi-mode (bi-mode) train switches between AC overhead electric mode and self-powered (diesel) mode. This happens at designated changeover locations.

Automatic Power Changeover (APCo)
APCo is operated automatically by a balise on the track. The balise signals the train to initiate the changeover automatically. If the APCo balise is defective, the signaller will tell you and you must carry out the PCO **manually** at the correct location.

Manual Power Changeover

Direction	Action at PCO Board
Diesel to Electric (D2E)	PCO to electric mode must be initiated at the "Warning of PCO" board. Place CPBC to OFF before initiating in-cab changeover.
Electric to Diesel (E2D)	PCO to diesel mode must be initiated at the "Warning of PCO" board. Place CPBC to OFF before initiating in-cab changeover.

Line Blocked to Electric Trains (§11.3)

If a line is blocked to electric trains and your train needs to be assisted through, you must:

- Lower all pantographs before reaching the "lower pantograph" sign
- Tell the driver of the assisting train when this has been done
- Keep all pantographs **lowered throughout** the movement
- Make sure APCo equipment on a multi-mode train is **disabled** throughout
- Disregard all lineside signs associated with a permanent traction changeover
- Not resume AC electric traction until passing the location the signaller has told you about, or the locations indicated by signage

Wrong-Direction Movements by Multi-Mode Train (§11.4)

If making a wrong-direction movement past a PCO location:

- Ensure APCo equipment is **disabled** throughout
- If starting in electric mode: lower all pantographs **before** reaching the end of OLE
- If starting in own traction power: do not raise pantographs until all are **beyond the start of OLE**

Source: GERT8000-AC §11.2–11.4, §18.3

PART 2 — RESPONDING TO A NEAR MISS

Source: EMR PPT 33-01 (Responding to a Near Miss)

11. WHAT IS A NEAR MISS?

Definition

"A near miss is an event that requires one or more people involved to make an **evasive or defensive action** in order to prevent an incident that could affect the safety of the train or the other person involved."

If without evasive/defensive action by either party there was potential for someone to be **injured, killed or property damaged** — it must be classed and reported as a near miss.

Source: EMR PPT 33-01 slide 3

Common Types of Near Miss

- Almost colliding with another vehicle, buffer stops or a person
- Almost striking a member of track staff on a running line
- Level crossing incident caused by a road user not adhering to crossing instructions, resulting in the train nearly colliding with a road vehicle

Source: EMR PPT 33-01 slide 5

12. IMMEDIATE ACTIONS AFTER A NEAR MISS

Source: EMR PPT 33-01 slides 6–7

Your initial reaction may be to apply the emergency brake to stop as soon as possible. This is not necessarily wrong, but remember:

Where to Stop

If the emergency brake is applied, the train will come to a stand and you will have **no control over exactly where it stops**.
If you still have control over the brakes you can control where the train actually stops.
Avoid stopping: in a tunnel, on a viaduct, at any other unsuitable place.

Once Stopped — Procedures to Follow

1. **Report the incident to the signaller immediately** using the correct radio procedure to ensure it is received as an emergency call.
2. Speak calmly and clearly. Act on the signaller's instructions.
3. The signaller will make subsequent trains aware of the conditions.
4. If there is a guard/conductor/on-board manager on the train, also report the near miss to them so they can deal with any affected passengers.
5. You will need to **complete a written report** to your local manager later.

13. FITNESS TO CONTINUE AFTER A NEAR MISS

Source: EMR PPT 33-01 slides 8–9

After a near miss you will likely experience the effects of adrenaline — this can make you feel alert and focused, or alternatively disorientated and unaware of your surroundings. Either way, allow **sufficient time for the effects to wear off** before deciding whether you are fit to continue.

Self-Assessment — Fitness to Continue

Ask yourself:
- Can I focus on the task at hand?
- Am I in a calm and logical state of mind?
- Am I shaken up by the incident?

Only continue if you are satisfied on all three points.

If You Decide You Are Not Fit to Continue

Inform the signaller immediately, then **notify control** so they can send another driver to relieve you.

No other party is permitted to judge you on this decision or force/influence your decision not to proceed.

Your safety and personal wellbeing takes priority over all else.

When your manager becomes aware of an incident, they will ensure you are met at the next appropriate location by a **competent person** who will ascertain your fitness to continue. If fit, you will be allowed to continue — unless the incident is too serious or the competent person has concerns.

14. CHAIN OF CARE

Source: EMR PPT 33-01 slide 10

Welfare Support Available After a Near Miss

Your manager may see you within **a couple of days** to check for any delayed reaction.

Transport home may be provided where considered necessary for your welfare.

You may be offered the opportunity to attend **debriefing sessions** with a suitable trained counsellor — arranged by your manager as soon as possible after the incident.

If you are **off sick** following the incident, your manager will carry out regular welfare visits and may get additional support from an occupational health provider.

PART 3 — OBSTRUCTIONS ON THE LINE

Source: GERT8000-TW1 Iss 22 §45 | GERT8000-M1 | EMR PPT 27-01

15. TYPES OF OBSTRUCTION

Source: EMR PPT 27-01 slides 5–6

You must follow the "trains put in danger" procedures (Section 16) if you see any of the following:

Obstructions on or near the Line

- Road vehicles on the line
- Road traffic collisions at level crossings
- Road over rail bridge hit by a road vehicle — debris on the line
- Cranes or similar plant collapsing and blocking the line
- Buildings being demolished falling on the line in error
- Large animals, landslides
- Items thrown or fallen on the track (shopping trolleys, pushchairs, fallen trees, snow drifts)
- Trespassers or fatalities
- Goods fallen off freight trains
- A train derailment or other rail accident

Damage to Structures or Earthworks

- Washouts, landslips, embankment failures, retaining wall failures
- Any **pooling of water** that may affect structures or earthworks
- Water rising up from the track or cess
- Unusual amounts of water pooling next to the track or in the cess
- Water flowing down or pouring out of the sides of embankments or cuttings

Cow or Bull — MUST Report

You must report a **cow, bull or other large animal** within the boundary fence to the signaller — even if it is not an immediate danger to trains. (TW1 §45.1)

16. WHEN OTHER TRAINS ARE PUT IN DANGER

Source: GERT8000-TW1 §45.1 | EMR PPT 27-01 slides 8–15

You must carry out the following if you see: an obstruction that could cause danger to other trains; damage to structures or earthworks; a cow, bull or large animal within the boundary fence; or something wrong with another train.

First Action — Tell the Signaller

Make a **Railway Emergency Call (REC)** on the train radio.
If you cannot make a REC — use any other means of direct communication from within the driving cab.

A — Trains in the OPPOSITE Direction Put in Danger

Step-by-Step Procedure

1. Warn any approaching driver: **sound a long blast of the horn** and switch on **hazard warning indication**.
   If you cannot switch on hazard warning indication — show a **red light forward**.
2. Continue for at least **1¼ miles (2 km) beyond the obstruction** and stop there.
   Do NOT pass any signal at danger unless already authorised to do so.
3. Tell the signaller by REC (from cab, or any other cab on the train, or other means).
4. **Stay at that location** until the signaller confirms:
   - Each driver whose train must remain at a stand has received and understood the message, OR
   - No train is approaching the obstruction.
5. Place a **TCOC (track-circuit operating clip)** on each affected line if the signaller tells you to.
6. Then continue on your journey.

If You Cannot Contact the Signaller After Stopping

Place a **TCOC on each affected line**.
Then continue to a location where you can contact the signaller by:
- Any means of direct communication
- Any lineside telephone
- Going to the signal box

If Signaller Says a Train IS Approaching

Remain at that location.
Place **three detonators, 20 metres (approx. 20 yards) apart**, on the line.
Show a **hand danger signal** to warn that train.

B — Trains in the SAME Direction Put in Danger

Step-by-Step Procedure

1. **Stop your train as soon as you can.** If not yet spoken to signaller, stop in a location from which you may be able to contact them.
2. Place a **TCOC on any affected line**.
3. Tell the signaller by REC (from cab, other cab, or other means from outside the cab if necessary, or lineside phone).

4. **Stay at that location** until the signaller says no train is approaching.
5. Then continue on your journey.

If Signaller Says a Train IS Approaching

Proceed **on foot towards that train**.
Show a **hand danger signal**.
Place **three detonators, 20 metres (approx. 20 yards) apart**, on the line.

If You Still Cannot Contact the Signaller

Switch on **hazard warning indication at the rear** of the train.
Tell the guard you are leaving the train.
Proceed **on foot in the direction trains will approach**.
Show a hand danger signal visible to approaching drivers.
On reaching a telephone or signal box: place **three detonators 20 metres apart** on the line BEFORE you speak to the signaller.

C — When a FOLLOWING Train Is Put in Danger (TW1 §45.2)

Following Train Rule

If you see an obstruction or something wrong that could put a following train in danger, you must **not proceed beyond the next stop signal** until you have told the signaller.

17. HITTING DETONATORS

Source: GERT8000-TW1 §13.1 | EMR PPT 27-01 slide 17

Exploding Detonators — You Must

Stop the train immediately (at a safe location if possible).
Make a **REC** and tell the signaller what has happened.
If you cannot contact the signaller directly, proceed at caution towards any obstruction, signal or hand signal.
Do not proceed beyond the next stop signal until you have spoken to the signaller.

18. RECEIVING AN UNEXPECTED HAND DANGER SIGNAL

Source: GERT8000-TW1 §13.2 | EMR PPT 27-01 slide 18

Procedure

If you observe a hand danger signal shown unexpectedly on a running line and you are NOT already stopping as a result of a REC:

1. **Stop your train immediately.**
2. Make a **REC** and tell the signaller what you have seen.
3. **Do not proceed** until given permission.
4. If stopped and able to speak to the person showing the signal — **find out why** and tell the signaller what has happened.

19. TRACK DEFECTS

Source: GERT8000-M1 | EMR PPT 27-01 slides 20–21

If you believe there is a track defect, tell the signaller as soon as possible. Include:

- The **location** of the defect
- The **type** of defect (using the terms in the table below)
- Whether there is a **bridge or viaduct** at or close to the location
- As much information as you can about the defect

Type of Defect	Description
A track defect that is SEEN	You can definitely see a broken rail, defective rail or broken fishplates
A track defect that is FELT	Unusual movement — lurch/dip, shaking/vibrating, pitch
A track defect that is HEARD	Unusual noise — bang, rattle, grinding
Deterioration of ride quality	You notice worse ride quality at a location compared to before — do NOT report to signaller; report to your Operator's Control at earliest opportunity

Other Reasons to Examine the Line

Broken, defective or distorted rails
Obstructions on the line
Flood, defective signalling equipment, bridge strikes
Persons on or near the line
Removal of a disabled train or train division
Damaged OLE, tunnel damage, other infrastructure defects

20. EXAMINING THE LINE — SPEEDS AND CONDITIONS

Source: GERT8000-M1 | EMR PPT 27-01 slides 23–27

When instructed by the signaller to examine the line, you must proceed at caution. Key rules:

Situation	Maximum Speed
General — examine OLE or any obstruction	20 mph (30 km/h)
Suspected track defect	20 mph over the affected portion
Line within a tunnel	10 mph through the tunnel
Overline bridge struck by road vehicle (once checked safe to pass)	5 mph under the bridge

Cannot Examine During

Darkness or poor visibility — unless a portable headlight is being used.
Through a tunnel — if the headlight has failed (unless a portable one is being used).
In all three of these cases you MUST be accompanied by the guard or other **competent person** (if one is immediately available) — or for OLE examination, a competent person at all times in these conditions.

Nothing Found

If nothing untoward is found, you must only inform the signaller the line is clear **once you have passed through and are clear** of the affected section.
Note: if trespassers were on the line, they could have moved to a different location by the time you reach them.

Obstruction Found During Examination

Bring your train to a stand immediately.
Report your findings to the signaller.
If you can move the obstruction and consider it safe and practical to do so, you may do so.

21. TRESPASSERS

Source: GERT8000-TW1 §46 | EMR PPT 27-01 slides 31–32

If you see any trespassers, report this to the signaller **immediately** using the train radio. Give as much information as possible including:

- Train reporting number
- Location of the trespassers
- Direction the trespassers are heading (if known)
- Description of the trespassers (number, age if apparent, etc.)

What the Signaller Will Do

Stop your train if it would proceed over the affected portion of line.
Tell you what is happening and to **proceed at caution** past the location.
If trespassers are likely to endanger trains: stop your train, then tell you to proceed at caution and **not to exceed 10 mph (15 km/h) through any tunnel**.

If Trespassers Are No Longer on the Line

If you find that the trespassers are no longer on or near the line, or do not appear to be in danger from approaching trains, you must **tell the signaller**.

Passengers on Train — What to Do

It is generally accepted that it is safer for people to remain on the train.
On a passenger train, tell the on-board staff or make a PA announcement.
If the train has a guard, they will contact you. Agree with the guard whether they should:

- Assist you in protecting the incident
- Stay with the train and reassure the passengers
- Take over your duties if you are unable, including using the radio emergency call facility

QUICK REFERENCE SUMMARY

Key Numbers and Distances

Fact	Value
Minimum distance from live OLE	2.75 metres (9 feet)
Speed limit with ADD isolated (pantograph raised)	Max 100 mph (160 km/h)
Speed for examining OLE or line (general)	Not to exceed 20 mph (30 km/h)
Speed through tunnel when examining line	Not to exceed 10 mph
Speed under bridge struck by road vehicle	Not to exceed 5 mph
Distance to continue past obstruction (opposite trains)	At least 1¼ miles (2 km)
Spacing of detonators	20 metres (approx. 20 yards) apart
Number of detonators placed	3 detonators
Max obstruction height to allow coasting	Not more than 150 mm (6 inches) below contact wire
Max consecutive droppers off to allow coasting	Not more than 2
Minimum OLE structures from tunnel/overbridge to allow coasting	More than 5
Coasting speed (20 mph restriction)	Not to exceed 20 mph through affected area

Procedure Mnemonics

Mnemonic	Stands For
DROP STOP BOX (ADD)	Pantograph drops automatically — STOP train — BOX (report by REC)
REC order	Railway Emergency Call — cab first, other cab second, other means third
TCOC	Track Circuit Operating Clip — placed on affected line to warn signaller of obstruction

Q & A — TEST YOURSELF

All answers sourced from GERT8000-AC Iss 10 and GERT8000-TW1 Iss 22

Part 1 — AC Electrified Lines

Q1. What is the minimum safe distance you must keep from live OLE?

Source: GERT8000-AC §3.1

Answer

2.75 metres (9 feet). This applies to you and anything you are carrying. To enter within this distance for any reason other than an emergency rescue, you must request the OLE is isolated and earthed.

Q2. What three things happen automatically when the ADD operates?

Source: GERT8000-AC §12.1b | EMR PPT 100-02 slide 14

Answer

1. The pantograph automatically lowers. 2. The VCB (Vacuum Circuit Breaker) opens. 3. The line light extinguishes.

Q3. What is the maximum speed if you proceed with an isolated ADD and the affected pantograph raised?

Source: GERT8000-AC §12.8

Answer

100 mph (160 km/h), until the pantograph has been examined and the ADD reset. You may be instructed to keep to a lower speed for some or all of the journey.

Q4. List ALL conditions under which you can continue to a suitable location (rather than stop immediately) when the line light goes out.

Source: GERT8000-AC §12.2

Answer

ALL of the following must apply: (1) Only one pantograph in use and it is on one of the first three vehicles. (2) The ADD is not isolated and has not operated. (3) No unusual movement of, or noises from, the OLE. (4) You have made one attempt to reset, which was not successful.

Q5. Name five possible actions when stopped in a neutral section (OHNS).

Source: GERT8000-AC | EMR PPT 100-02 slide 11

Answer

1. Reset/close the VCB — your pantograph may not be in the dead section. 2. Arrange for assistance and roll the train out of the OHNS. 3. Raise another pantograph on the train. 4. Change to diesel/self-powered mode on a multi-mode train. 5. Lower all pantographs and arrange for the train to be assisted through (as for a line blocked to electric trains).

Q6. What must you check before leaving your train to examine the OLE?

Source: GERT8000-AC §12.5

Answer

Check there is no damaged OLE in the vicinity of the door you would use to exit. If there is, you must stay in the train and tell the signaller. The signaller must also tell you when you can leave.

Q7. What four conditions must be met before you can confirm it is safe to coast with pantographs lowered?

Source: GERT8000-AC §14.2

Answer

(1) Any obstruction is not more than 150 mm (6 inches) below the contact wire. (2) Not more than two consecutive droppers have come off. (3) The object or defect is more than FIVE OLE structures away from a tunnel or overbridge. (4) No other defect is obvious.

Q8. A high-speed coasting advance warning board has its white flashing lights defective. What must you do?

Source: GERT8000-AC §15.5

Answer

Report this to the signaller after passing through the affected area, stopping specially to do so.

Q9. You are detained with a damaged pantograph that is not severe. Can you raise it? What rules apply?

Source: GERT8000-AC §12.7

Answer

Yes — you may raise it to supply electrical power (heating, lighting) while waiting for assistance. Rules: (1) Immediately after raising, check it is correctly in contact with OLE and there is no arcing. (2) No movement of the train is allowed with the pantograph raised. (3) The pantograph must be lowered before the assisting train is attached.

Q10. When a line is blocked to electric trains and your multi-mode train is assisted through, what must you do about APCo?

Source: GERT8000-AC §11.3

Answer

Make sure the APCo equipment is disabled so it will not operate throughout the movement — otherwise it may automatically raise a pantograph.

Part 2 — Near Misses

Q11. How is a "near miss" defined?

Source: EMR PPT 33-01 slide 3

Answer

An event that requires one or more people involved to make an evasive or defensive action in order to prevent an incident that could affect the safety of the train or the other person involved. If without evasive or defensive action there was potential for someone to be injured, killed or property damaged, it must be reported as a near miss.

Q12. You decide you are not fit to continue after a near miss. What must you do, and can anyone overrule you?

Source: EMR PPT 33-01 slide 9

Answer

Inform the signaller immediately, then notify control so they can send another driver to relieve you. No other party is permitted to judge you on this decision or force or influence your decision not to proceed. Your safety and personal wellbeing takes priority over all else.

Part 3 — Obstructions on the Line

Q13. You see a cow inside the boundary fence but not on the line. Do you need to report it?

Source: GERT8000-TW1 §45.1

Answer

Yes. You must report a cow, bull or other large animal within the boundary fence to the signaller by REC — even if it is not an immediate danger to trains.

Q14. Trains in the opposite direction are put in danger. How far beyond the obstruction must you travel and stop?

Source: GERT8000-TW1 §45.1a

Answer

At least 1 and a quarter miles (2 km) beyond the obstruction. You must not pass any signal at danger unless already authorised to do so.

Q15. How many detonators do you place and how far apart?

Source: GERT8000-TW1 §45.1

Answer

Three detonators, placed 20 metres (approximately 20 yards) apart on the line.

Q16. Trains in the same direction are put in danger and the signaller says a train is approaching. What do you do?

Source: GERT8000-TW1 §45.1b

Answer

Proceed on foot towards that train, show a hand danger signal, and place three detonators 20 metres apart on the line.

Q17. Your train explodes a detonator. What must you do?

Source: GERT8000-TW1 §13.1

Answer

Stop the train immediately. Make a REC and tell the signaller what has happened. If you cannot contact the signaller, proceed at caution towards any obstruction, signal or hand signal. Do not proceed beyond the next stop signal until you have spoken to the signaller.

Q18. You receive an unexpected hand danger signal and are not already stopping from a REC. What must you do?

Source: GERT8000-TW1 §13.2

Answer

Stop the train immediately. Make a REC and tell the signaller what you have seen. Do not proceed until given permission. If stopped and able to speak to the person showing the signal, find out why — and tell the signaller what has happened.

Q19. What is the maximum speed when examining the line through a tunnel?

Source: GERT8000-M1 | EMR PPT 27-01 slide 26

Answer

10 mph. You must not exceed 10 mph through the tunnel.

Q20. You are examining the line after a road vehicle struck an overline bridge. You have checked it is safe to pass. What speed limit applies under the bridge?

Source: GERT8000-M1 | EMR PPT 27-01 slide 26

Answer

5 mph. You must not exceed 5 mph under the bridge.

Q21. You notice ride quality has deteriorated at a particular location. Must you report this to the signaller?

Source: GERT8000-M1 | EMR PPT 27-01 slide 21

Answer

No. Deterioration of ride quality does not need to be reported to the signaller. You must report it to your Operator's Control at the earliest opportunity.

Q22. You see trespassers and the signaller says they are likely to endanger trains. What speed limit applies through a tunnel?

Source: GERT8000-TW1 §46 | EMR PPT 27-01 slide 32

Answer

10 mph (15 km/h) through the tunnel.

Q23. You have examined the line and found nothing wrong. When can you tell the signaller the line is clear?

Source: GERT8000-M1 | EMR PPT 27-01 slide 29

Answer

Only once you have passed through and are clear of the affected section of line — not before.